

Performance Comparison of Single- and Dual-Camera Setups for Image-Based Satellite Orbit Determination

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Abstract

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1 Introduction

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2 Methods

2.1 Single-Camera orbit determination

2.1.1 Gaussian Method

The Gaussian method's approach is to numerically fit observations to a physically feasible orbit. The algorithm yields parameters for full orbit characterization, namely semi major axis a , eccentricity e , argument of periapsis ω and plane normal vector n .

A single observer camera is required, whereas three measurements each consisting of position vector p_i , view vector v_i and corresponding timestamp t_i need to be recorded to determine a solution. [1]

2.1.2 Simplified Gaussian Method

The simplified Gaussian method is a self developed approach to numerically fit observations to a physically feasible circular orbit. Its incentive stems from the condition that satellite orbits low above earth's surface possess very low eccentricities and can mostly be considered circular. The algorithm yields parameters for circle-orbit characterization, namely semi major axis a and plane normal vector n .

Similar to the Gaussian method, a single observer camera is required, whereas only two measurements each consisting of position vector p_i , view vector v_i and corresponding timestamp t_i need to be recorded to determine a solution. Hereby, the solution consists of two scalar parameters ρ_1 and ρ_2 which must satisfy the radius equality condition as well as the arc of a circle condition stated as follows:

$$|r_1| = |r_2| \Leftrightarrow \sqrt{(p_1 + \rho_1 v_1)^2} = \sqrt{(p_2 + \rho_2 v_2)^2} \quad (1)$$

$$\arccos\left(\frac{(p_1 + \rho_1 v_1) \cdot (p_2 + \rho_2 v_2)}{|(p_1 + \rho_1 v_1)|^2}\right) \cdot |(p_1 + \rho_1 v_1)|^{\frac{3}{2}} = (t_2 - t_1)\sqrt{\mu} \quad (2)$$

2.2 Two camera trajectory determination

2.2.1 Full parametric trajectory

2.2.2 Circular parametric trajectory

2.3 Observation setup

2.4 Parameter stability definition

2.5 Error propagation definition

3 Results and Discussion

3.1 Parameter stability

3.2 Error propagation

3.3 Lower altitude bound for feasible trajectory determination

3.4 Stability of circular models applied to elliptical orbits

4 Conclusion

References

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